

Pure states as real extreme points

`MathlibAnnex.Analysis.CStarAlgebra.IsPureState`

def

Defines purity inside the normalized positive state space.

Statement

Let A be a unital complex C^* -algebra, and let $\varphi : A \rightarrow \mathbb{C}$ be continuous and complex-linear. Purity is the following condition on φ .

Definition

Write

$$S(A) = \{\psi : A \rightarrow \mathbb{C} \text{ continuous and complex-linear} : \psi(a) \geq 0 \text{ for all } a \geq 0, \psi(1) = 1\}.$$

Then φ is pure precisely when $\varphi \in S(A)$ and, whenever $\psi, \eta \in S(A)$ and $0 < t < 1$ satisfy

$$\varphi = (1 - t)\psi + t\eta,$$

one has $\psi = \eta = \varphi$. Equivalently, $\varphi \in \text{ext}_{\mathbb{R}} S(A)$. Membership in the extreme-point set already includes membership in $S(A)$.

Assumptions

The definition does not require $A \neq 0$. Positivity uses the given star-compatible order on A and the usual real nonnegative cone inside $\mathbb{C}$.

Conclusion

A pure state is in particular a state; extremality is taken for real convex combinations.

Main citations

- Exact definition and context

Lean source signature (exact)

The complete declaration below is a separate exact source excerpt; the original header record is retained with the manuscript.

```
-- A pure state is an extreme point of the ordinary state space. -/
def IsPureState (A : Type u) [CStarAlgebra A] [PartialOrder A] [StarOrderedRing A]
```

```
(phi : A → L[ℂ] ℂ) : Prop :=
phi ∈ (stateSpace A).extremePoints ℝ
```

In the source	Mathematical meaning
(A : Type u) [CStarAlgebra A]	The unital complex C^* -algebra on which the functional is defined.
[PartialOrder A] [StarOrderedRing A]	These typeclasses provide the usual order on a C^* -algebra, equivalently $a \leq b$ when $b - a$ is positive. They are used by the state and positivity arguments.
(phi : A → L[ℂ] ℂ)	The given continuous complex-linear functional φ .
: Prop	The output is a property of that functional.
stateSpace A	The set $S(A)$ of all positive normalized continuous complex-linear functionals defined above.
(stateSpace A).extremePoints ℝ	The real extreme points of that convex set: no nontrivial convex decomposition into two different states.
phi ∈ (stateSpace A).extremePoints ℝ	The entire RHS says both that φ is a state and that every decomposition displayed above has $\psi = \eta = \varphi$.

Exact source and proof.

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▼ Exact content identity

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